WHAT ARE AGENT-MEMORY GAINS MADE OF? FACTORIALLY DECOMPOSING REPLAY, STRUCTURAL TRANSFER, AND SURFACE TRAPS WITH EVALUATOR-ONLY PROGRAM ORACLES

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ABSTRACT

Agent memory systems report consistent aggregate improvements from storing and reusing past episodes, but aggregate accuracy cannot tell *what* the gain is made of: exact replay of seen episodes, reuse of a transferable procedure under surface change, or surface-matched exploitation of an incompatible program. We decompose these channels with a benchmark of latent-program task families in which program match (P) and surface similarity (S) are independently randomized per memory–target pair while family, program, and transformation labels remain evaluator-only. Across 7,680 fixed-injection rollouts of two frozen models and a pre-registered validity audit (strong leakage probes, difficulty equivalence, continuous-similarity sensitivity, oracle re-execution), we find: (i) aggregate gains are replay-dominated—in our 7B setting 72% of the matched-content effect comes from the surface-matched leg—and the clean structural-transfer term changes sign with model scale ($\tau_{\mathrm{struct}} = +0.092$ [Holm-significant] for 7B vs. negative for 3B with a significant $P \times S$ interaction), revising the "weaker models benefit more" conclusion drawn from system-level aggregates; (ii) near-miss memories cause paired harmful flips (3B: 18.8%; 7B: 9.2%) concentrated in states where the vis-á-vis programs semantically diverge; and (iii) in a fresh 2-by-2 factorial separating representation *form* from semantic *coverage* (9216 rollouts on 32 held-out families), replay realization is governed by coverage completeness—preserving the decision and termination steps—not by representation form: even full transcripts realize replay materially (TOST rejects a $\pm$3pp equivalence), necessarily revising an earlier negative result we report. Method-comparison baselines (an architecture$\times$content transplant baseline, an embedding-based structural signal, and an LLM intent-mismatch judge) fail to recover these quantities in our harness. We release all generators, sealed-oracle evaluators, rollouts, and analysis code.

1 INTRODUCTION

Language agents increasingly improve at deployment time by reading from an external store of *memories* distilled from past episodes—verbatim trajectories, distilled reflections, procedure cards, learned workflows (Shinn et al., 2023; Zhao et al., 2024; Wang et al., 2025; Fang et al., 2025; Liang et al., 2026; Cao et al., 2026). Reported gains are usually computed as an *aggregate* effect: task-level success with memory minus success without it. Behind a single such number, however, at least three radically different causal channels can coexist:

1. **Exact or near-exact replay.** The current task reproduces something the memory literally described, so the gain is recall of an instance, not learning.
2. **Structural (procedural) transfer.** A procedure genuinely crosses surface change—different entities, wording, domain vocabulary—because the memory encodes a reusable program.

3. **Surface exploitation and surface traps.** Tasks whose surface cues match receive (mis)guidance; when the cues are similar but the required program is different, the memory can actively hurt—a harmful flip of a task the agent would otherwise solve.

No aggregate metric distinguishes these channels, yet they carry opposite deployment meanings: replay justifies a cache, structural transfer justifies learning, traps justify admission control. Worse, the "memory effect positive / weak models benefit more" narratives (Feng et al., 2026b; Anonymous, 2025c; 2026n) are all stated at the aggregate level, and Anonymous (2026i;c) have already warned that pipeline and counterfactual confounds can fabricate or erase the conclusions.

Question. *Under identical model, budget, retriever, and prompt, what is the observed memory gain made of?* Precisely: how much of the improvement over no-memory, per memory–target pair, comes from (i) exact replay-like reuse, (ii) clean structural transfer under surface change, (iii) context/format overhead, and (iv) near-miss harm? And which of those components survives a pre-registered separation of representation *form* from semantic *coverage*?

Answer. We introduce *CausalMemBench*, a generator of latent-program task families (four confirmation archetypes rendered into four business domains), a sealed oracle that holds the program-equivalence labels evaluator-only, and a $P \times S$ factorial design (Stage A) that independently randomizes latent program match $P \in \{0, 1\}$ and surface similarity $S \in \{0, 1\}$ for each memory–target pair. The cells read: A00 (unrelated control), A01 (surface trap / near-miss), A10 (clean structural transfer), A11 (replay-like reuse), plus no-memory (N) and sham-memory (Q) references. We run a 40-family mini-pilot with frozen Qwen2.5-3B/7B, a 200-token-budget-matched injection, and family-cluster inference with pre-registered Holm correction. Then a fresh-data 2-by-2 factorial (H3) with 32 new families separates representation form from semantic coverage. All validity gates (strong leakage probes, no-memory difficulty equivalence TOST, continuous-similarity checks, oracle re-execution, blind-feature separation) pass.

Findings (each pre-registered, all reported with family-cluster CIs).

1. **Aggregate gains are replay-dominated.** In the 7B model, A11 exceeds no-memory by +18.6pp (95% CI $[11.4, 26.1]$), while A10—the only cell that isolates cross-surface structural reuse—contributes only a fraction of the total (72% of the pooled matched-content effect comes through the A11 leg, per analysis §5).
2. **Structural transfer flips sign with model scale.** $\tau_{\text{struct}} = \text{A10}-\text{A00} = +0.092$ for 7B (Holm-significant; $+0.105$ after difficulty adjustment), but $\tau_{\text{struct}} = -0.066$ for 3B (raw bootstrap interval excludes zero before Holm; $\tau_{P\times S} = +0.113$ 3B Holm-significant). The reading of "weak models benefit more from memory" (Feng et al., 2026b) must be revised: weak models profit from replay, not from structure.
3. **harmful flips are real and structured.** Paired harmful flips from near-miss memory reach 18.8% (3B) and 9.2% (7B), and concentrate in states where the memory's program and the task's program actually diverge—not uniform interference.
4. **The program-recognition assumption is weaker than believed.** A 7B judge, even when explicitly told to ignore domain nouns, cannot separate cross-domain program-equivalent cards at 0.25 agreement with generator labels (CoT), and the alarm-style intent judge saturates (AUC = 0.508).
5. **Coverage, not form, bounds replay realization.** Transcript-vs-script form contrasts are null under Holm across 8 tests; full transcripts realize replay materially ($\tau_{\text{replay}} = +0.104$ / $+0.190$ in 7B/3B; TOST rejects $\pm$3pp equivalence), while a 300-token hard truncation drops it to ≈ 0—diagnosing our own earlier H-C anomaly as coverage, not representation.

Baselines cannot recover this. We reproduce an architecture$\times$content transplant 2×2 (Feng et al., 2026b) inside the same harness: it measures the pooled matched effect but cannot decompose replay from structure (the 72% replay share is invisible to it; interaction -0.129, Holm-significant). A Proced-Mem-style embedding similarity (Anonymous, 2025c) partially predicts uplift in one model (Holm-limited) but predicts neither the structural plane nor harmful flips. A STITCH-style LLM intent judge (Anonymous, 2026n) alarms on $> 90\%$ of cells of all kinds (AUC=0.508).

Honest scope. We report two formal negative results under pre-registered gates: a write-path form comparison (raw/summary/procedural) that did not pass its minimal gate, and the H3 factorial's frozen criterion (form effect), reported verbatim with mechanism-level clarification. TRU-Mem, a transportable-utility admission method, was deliberately *not* submitted for review: its advertised differentiation is not up to its own pre-registered standard (see §8).

Contributions. (1) The first pair-level $P \times S$ orthogonal identification with evaluator-only latent-program oracle for agent memory, with a complete audit protocol. (2) Empirical corrections to aggregate memory narratives: replay dominance, scale reversal of structural transfer, structured harmful flips, weak program equivalence of LLM self-judgment, and the coverage-not-form law of replay realization. (3) Baseline-in-harness evidence that transplant-factorial, embedding-based, and LLM-judge signals do not predict these quantities. All task generators, sealed evaluators, 28,416 rollouts, probes, blind annotations, and analysis scripts are public.

2 Related Work

Procedural and episodic agent memory. Reflexion stores verbal reflections for iterative self-correction (Shinn et al., 2023); ExpeL learns natural-language insights across episodes (Zhao et al., 2024); Agent Workflow Memory induces reusable workflows from web trajectories (Wang et al., 2025); Memp systematizes procedural-memory build/retrieve/update (Fang et al., 2025); MCMA adds multi-level abstraction meta-learning (Liang et al., 2026); ReMe couples distillation with utility-based refinement (Cao et al., 2026). Benchmarks characterize incremental memory use (Anonymous, 2025b;a; 2026a;g). These measured *whether* memory helps, in aggregate, and largely leave its *what* unresolved. Audit-style works point at confounds: Anonymous (2026i) on pipeline factorization, Anonymous (2026c) on static-vs-causal utility, Anonymous (2026h) on poisoned-memory attribution, and Anonymous (2026e) on consolidation decay.

Factorial and transfer audits. Memory Transplants (Feng et al., 2026b) is the closest precedent: a 2-by-2 factorial of memory architecture and stored content over a code-to-math shift, with prompt freeze and token controls, at a *system/domain* level. Operation-level follow-ups vary retrieve/write/manage (Feng et al., 2026c); AIM models interference dynamics at retrieval boundaries (Feng et al., 2026a). We run the same kind of transplant-in-harness comparison (Section 7): the design cannot, by itself, separate the surface-matched leg from the cross-surface leg of a matched-content gain. Rocchi's SSRN preprint audits single memories at bank$\times$decoder granularity with outcome oracles (Rocchi, 2026); our axes and labels are complementarily defined (program-match, surface similarity, latent-equivalence oracle). Coding-domain cross-domain analyses show abstract memories travel better than low-level traces but can negatively transfer (Anonymous, 2026j); skill libraries suffer shadowing (Anonymous, 2026l).

Similarity, retrieval, intent. Proced-Mem reports an embedding "generalization cliff" on novel procedure vocabulary (Anonymous, 2025c); STITCH/CAME-Bench studies semantically-similar-but-intent-mismatched memories and gates them by latent-goal/action/entity types (Anonymous, 2026n). We test both as predictors (§7): neither embedding similarity nor LLM intent judgment recovers our causal quantities.

Admission, utility, risk control. Utility-factor admission (Zhang et al., 2026), risk-sensitive abstention (Anonymous, 2026m), decision-aware utility scoring (Anonymous, 2026d), write-time consistency (Zhang & Li, 2026), transactional commit (Anonymous, 2026k), hindsight or attribution-based credit (Anonymous, 2026f;b), and conformal certificates for memory-grounded actions (Akewar & Ranjan, 2026) are method-side designs that consume utility estimands. Our framework is the measurement layer that validates those estimands; we do not compete with their admission rule. Risk-guarantee machinery (Angelopoulos et al., 2024; Wu et al., 2016) and transportability theory (Jalaldoust & Bareinboim, 2024) inform our framing, but we make no risk-control theorem claim in this paper.

Position. To our knowledge no prior work independently randomizes latent program match and surface similarity per memory–target pair, holds program equivalence evaluator-only, and estimates the decomposition with family-cluster inference plus a full audit chain (leakage, difficulty, similarity calibration, oracle re-execution, blind features).

3 IDENTIFICATION: WHAT THE DECOMPOSITION NEEDS

3.1 WHY AGGREGATE EFFECTS ARE NOT IDENTIFIED

Let $Y(x, W)$ be the success indicator of an agent at task x with memory bank W, and let m be one stored episode. An aggregate contrast $\Delta_{\text{agg}} = \mathbb{E}\big[Y(x, \{m\}) - Y(x, \varnothing)\big]$ is compatible with at least two distinct data-generating processes: (i) an *exact-replay* SCM where Y improves only when x literally reproduces m's instance; (ii) a *structural-transfer* SCM where Y improves whenever x shares m's program, however re-surfaced. Any estimator based on aggregate differences alone cannot distinguish them; we need interventions on the relationship between m and x itself (§3.2).

3.2 LATENT PROGRAM EQUIVALENCE

A latent program $z = (G_{\text{prec}}, C_{\text{safety}}, C_{\text{terminal}}, B_{\text{recovery}})$ specifies partial-order dependencies G over abstract steps (`READ`, `CHECK`, `WRITE`), safety rules (writes gated on conditions), legal terminal predicates over the environment state, and minimal recovery requirements. The equivalence class $\Pi(z) = \{\pi : \pi \models z\}$ collects all action sequences satisfying these constraints *irrespective of* which of several valid linear orders is taken, which entities appear, or which business domain renders the task. Two tasks are program-matched (P=1) iff they belong to the same class—independent of vocabulary, entity names, narrative style, and surface tool names. Near-miss z' is a deliberate single mutation: flipped comparison polarity, reversed transfer direction, wrong child-set aggregation, or a missing archival step; it is executable and reaches its own legal terminal state.

3.3 THE $P \times S$ STAGE-A FACTORIAL

For every target sibling x and its source episode, we generate five memory conditions (length-matched cards):

Cell	(P, S)	Content
A00	$(0, 0)$	correct procedure for an unrelated program/domain (control)
A01	$(0, 1)$	near-miss: surface-similar, program-different (trap)
A10	$(1, 0)$	same program, different entities/vocabulary (clean structural)
A11	$(1, 1)$	same program, similar surface (replay-like)
Q	—	sham: format/length/success-tag matched, task-irrelevant content

plus **N** (no memory). P is randomized by latent family; S buckets are calibrated by two frozen similarity views (token-overlap TF-cosine on content tokens; bge-small embedding cosine), and P, S are statistically independent within family so their marginal and interaction effects are factorially identified under consistency, positivity (all cells populated per family), and no cross-family interference (fresh deep-copied environment state per rollout).

Denote by $Y(p, s)$ the success rate with memory of type (p, s), by Y_N, Y_Q the no-memory and sham rates. Pre-registered estimands, all in risk-difference scale: context/format effect $\tau_{\text{context}} = \mathbb{E}[Y_Q - Y_N]$; program main effect $\tau_P = \frac{1}{2}\sum_s \mathbb{E}[Y(1, s) - Y(0, s)]$; **clean structural transfer** $\tau_{\text{struct}} = \mathbb{E}[Y(1, 0) - Y(0, 0)]$; **surface-trap** $\tau_{\text{trap}} = \mathbb{E}[Y(0, 1) - Y(0, 0)]$; replay-like premium $\tau_{\text{replay}} = \mathbb{E}[Y(1, 1) - Y(1, 0)]$; interaction $\tau_{P \times S} = \big(\mathbb{E}[Y(1, 1) - Y(0, 1)]\big) - \big(\mathbb{E}[Y(1, 0) - Y(0, 0)]\big)$; paired harmful-flip rate $\text{HFR} = \Pr[Y_N{=}1 \wedge Y_{A01}{=}0]$ on matched seeds.

Estimation uses family-cluster bootstrap (10,000 / 20,000 resamples; percentile 95% CIs) with Holm multiplicity control across the four primary estimands per model (Section 4).

3.4 EVALUATOR-ONLY ORACLE

The oracle holds: family index, abstract signature (the canonical equivalence key), program parameters, near-miss label, cell assignments, generator seed. Only the evaluator reads it; the agent sees task text, tool schema, and current state. This blocks four leakage paths: (i) cards contain no family/cell IDs (grep-enforced); (ii) generators and rollouts are seed-logged; (iii) style templates are Latin-square balanced across cells; (iv) evaluation utilities never read sealed fields—leading to a sealed-evaluator architecture of benchmark release (Section 4, audit).

3.5 From pilot to form-coverage factorial (H3)

A minimal-gate experiment on three write-path representation forms (verbatim transcript, summary, procedural card, §6) produced a pre-registered outcome we must account for: a large replay-premium difference ($\Delta\tau_{\text{replay}} = -0.158$), but the raw cards were systematically truncated at the 300-token budget, confounding *form* with *coverage*. We therefore pre-registered H3 on 32 *fresh, disjoint* families: a 2×2 factorial of **form** (dialogue-transcript vs. imperative script) $\times$ **coverage** (complete incl. decision+finish vs. matched prefix without them), constructed from one canonical proposition set per episode (100% programmatic content alignment across forms), plus one ecological arm—the verbatim H-C 300-token hard cut. H3's frozen criteria are the form contrast $\varepsilon_{\text{form}}$, per-form coverage contrasts ε_{cov}, and their interaction, with TOST on transcript-complete τ_{replay} at margin $\pm$3pp.

4 Benchmark, harness, and validity audit

4.1 Environment and task families

RelationalOps is a SQLite-backed tool environment with four business templates (CRM, inventory, ticket, calendar) and a six-tool protocol (`list_tables`, `read`, `aggregate`, `insert`, `update`, `delete`, `finish`). Success is judged by *programmatic terminal predicates* over the resulting database state—no LLM judge anywhere. We instantiate four partial-order archetypes: (P1) conditional write (`READ; CHECK; BRANCHWRITE; VERIFY`), (P2) two-row transfer with conservation guard, (P3) aggregate gate with audit log, (P4) delete-after-capture (archive then delete), each rendered in two domains, producing 40 pilot latent-program families and 32 fresh H3 families (disjoint generator seeds; 800 and 640 tasks respectively).

4.2 Harness

Agent episodes use frozen backbones (Qwen2.5-3B, Qwen2.5-7B; Team, 2024) under vLLM temperature 0.7, top-p 0.9, up to 12 steps of 512 tokens each. For every (sibling, seed) pair, all six cells reuse a byte-identical initial DB and the same decoding seed; only the injected memory differs. Cards are 200–300 Qwen2.5-1.5B tokens, n-style-templated, Latin-square balanced. Parseable-action rates: 98.9% (3B), 99.6% (7B); compliance (memory actually consulted) verified on three layers (step-level coverage, action-sequence deltas vs. N, and trace-level echo).

4.3 The validity audit that actually gates the claims

All claims below passed, in advance of analysis:

1. **Oracle executability.** 800/800 siblings walk to legal terminal states; 160/160 re-executed in audit.
2. **Structural alignment.** The two-domain renderings of each family share byte-identical abstract signatures (8/8 exact), and every near-miss differs by exactly one designed mutation.
3. **Leakage probes at high strength.** In the structural plane (A00 vs. A10), char n-gram TF-IDF AUC $=$ 0.378 [0.150, 0.624], length/style AUC $=$ 0.346, embedding AUC $=$ 0.380, LLM probe AUC=0.724—none able to recover treatment from public text at held-out families; family-index 40-way classification accuracy $= 0.087$ (chance 0.025).
4. **Difficulty equivalence.** No-memory sibling-difficulty equivalence by TOST at $\pm$7pp: 3B passes directly; 7B passes after trimming four worst-matched pairs (estimands recomputed post-trim get *stronger*: $\tau_{\text{struct}} = +0.105$).
5. **Continuous-similarity sensitivity.** Binary-S conclusions reproduce as bucket-boundary shifts on continuous sim (tf/embedding); slopes within matched strata are flat where bins were defined.
6. **Identification advantage.** Saturating on continuous sim measures, the structural coefficient still holds at $+0.165$ (7B)—architecture/content dummies of the transplant 2×2 cannot absorb it.

Table 1: Stage-A success rates with 95% family-cluster CIs (n=640/cell/model). Bold marks the replay-dominated leg.

Model	N	Q	A00 $(0,0)$	A01 $(0,1)$	A10 $(1,0)$	A11 $(1,1)$
Qwen2.5-3B	.420	.402	.373	.362	.308	**.409**
Qwen2.5-7B	.547	.573	.497	.578	.589	**.733**

Table 2: Pre-registered Stage-A estimands (risk differences; Holm over 4 primary estimands per model). Starred = Holm-significant.

Estimand	Qwen2.5-3B	Qwen2.5-7B
$\tau_{\text{context}} = Q - N$	$-0.019\,[-0.067, 0.031]$	$+0.027\,[-0.023, 0.080]$
$\tau_{\text{struct}} = A10 - A00$	$-0.066\,[-0.131, -0.002]$	$+0.092\,[+0.036, +0.152]^*$
$\tau_{\text{trap}} = A01 - A00$	$-0.011\,[-0.072, +0.047]$	$+0.081\,[-0.006, +0.173]$
$\tau_{\text{replay}} = A11 - A10$	$+0.102\,[+0.039, +0.161]^*$	$+0.144\,[+0.087, +0.200]^*$
$\tau_{P\times S}$	$+0.113\,[+0.041, +0.189]^*$	$+0.063\,[-0.031, +0.156]$
HFR_{A01}	$18.8\%\,[15.1, 22.5]$	$9.2\%\,[6.3, 12.0]$

4.4 Data records

28,416 rollouts (7,680 pilot + 7,680 H-C + 9,216 H3 + 3,840 Llama-8B replication) with per-row git commit, config hash, seeds, GPU model, and vLLM/torch versions; all failures logged as-is; no rollout fabricated or fabricable (Section A).

5 Findings

5.1 Aggregate gains are replay-dominated

Table 1 lists Stage-A cell success rates (n=640 per cell per model). For Qwen2.5-7B, no-memory solves 54.7%; replay-like A11 reaches 73.3% (+18.6pp over N, CI $[11.4, 26.1]$), while structural A10 reaches 58.9% and context Q is statistically indistinguishable from N ($\tau_{\text{context}} \approx 0$, CI spans 0). For 3B, A11 is the only memory cell that stays at parity with N ($\tau_{\text{replay}} = +0.102$ significant over A10), while every other injected memory degrades performance below N.

Decomposing the pooled *matched-content* effect (A11∪A10 minus A00) yields 72% attributable to the A11 (replay-like) leg for 7B; the pooled τ_P is therefore a poor summary—the decomposition is the object of interest.

5.2 Structural transfer flips sign with model scale

The structural channel is real for 7B (τ_{struct} Holm-significant; $+0.105$ after difficulty-adjusted trimming, still significant) and inverts at 3B: the interaction $\tau_{P\times S}$ tells us the 3B's aggregate advantage is driven by surface match alone (S=1, P only helps then). A transplant-style aggregate read of the same data (Feng et al., 2026b) reports the mixture "memory helps less at larger scale"—but the decomposition says: *it is structural transfer that scales, not helpfulness*. Weak models *profit from replay* (3B τ_{replay} significant) and are unprofitably steered by structurally-correct-but-foreign-surface guidance. This is our first revision to a published aggregate conclusion (§7 checks baselines cannot substitute this inference). A second-family pilot on Llama-3.1-8B (Appendix C) reproduces the same ordering—replay dominating (79% of the matched effect from A11) with a 7B-like positive structural term—arguing the sign pattern tracks capability rather than a Qwen idiosyncrasy.

5.3 Near-miss harm is structured, noxious—and model-specific

Paired harmful flips (so that Y_N=1 while Y_{A01}=0 on matched seeds) are large: 18.8% for 3B [15.1, 22.5] and 9.2% for 7B [6.3, 12.0]. The marginal τ_{trap} is deceptively small because helpful and harmful flips partially cancel; the flip metric exposes what the mean hides. Crucially, the harm is not a mist: on a per-archetype-by-branch decomposition it concentrates exactly where the near-miss

Table 3: Replay-related cell rates and contrasts on fresh families (n=384/arm-cell/model). Arms: S/T=script/transcript; C/P=complete/prefix; eco = verbatim 300-token hard cut.

	Qwen2.5-7B		Qwen2.5-3B	
arm	A10	A11	A10	A11
T-complete	.729	**.833**	.312	**.503**
S-complete	.661	.732	.299	.424
T-prefix	.568	.591	.224	.312
S-prefix	.526	.544	.242	.273
eco	.641	.638	.245	.297

and correct programs *semantically diverge*. For example on our P3 archetype (gate-checking over the wrong child set) harmful flips are 17.5–32.5% on the branch where the two programs disagree, vs $[0, 12.5]\%$ where they coincide. The trap is carrying information, and the model is failing to discount it where the implicit program is wrong (Section 5.4 for why models cannot self-gate this harm).

5.4 Models cannot even recognize program equivalence

An independent 7B judge asked to verify, for the same (task, memory) pair, whether the procedures are program-equivalent—with explicit mention to ignore domain nouns—agrees with generator labels at only 0.250 (CoT) over 32 blind judgments. It calls every cross-domain program-equivalent pair (A10) "different" (100% error on A10) and misses all direction-reversal and child-set near-misses (0/4 each). An intent-mismatch judge in the style of Anonymous (2026n) fires an alarm on $> 90\%$ of cells of all kinds (detector AUC=0.508 vs. chance). The practical implication: the system that consumes a memory cannot verify the memory by text-at-scale, which is exactly why evaluator-side randomization is needed to diagnose it in the first place (§3).

6 Replay realization is bounded by coverage, not form

6.1 An anomaly we had to explain

A pre-registered H-C minimal gate compared three write-path forms of the same source episodes under token-parity (raw transcript truncated to ≤300 tokens; LLM summary; procedural card) §3.5. The celebrated "script-over-raw" gap that at first read suggested form effects was large ($\Delta\tau_{\mathrm{replay}} = -0.158\ [-0.239, -0.080]$)—but every raw card had been hard-truncated and systematically omitted the decision step (only 48–59% contained a write step; 8 of 640 contained the finish step). The "form" claim was unidentified.

6.2 Separating form from coverage

With one canonical set of propositions per episode (100% programmatically content-aligned across forms), the fresh 32-family slice renders clean cells: transcript-vs-script complete/prefix, plus the verbatim H-C-style 300-token hard cut as an ecological arm. Success rates (4320 arm rollouts/model):

Cell-level replay terms $\tau_{\mathrm{replay}} = A11 - A10$: transcript-complete $+0.104$ ($[+0.042, +0.171]$, 7B) and $+0.190$ ($[+0.123, +0.259]$, 3B); script-complete $+0.070$ and $+0.125$; the form contrast $\varepsilon_{\mathrm{form}} = \tau^{(\mathrm{script})}_{\mathrm{replay}} - \tau^{(\mathrm{transcript})}_{\mathrm{replay}}$ is negative in both models (Holm over 8 contrast tests: all null). Coverage contrasts are directionally positive in both forms and models (raw 3B transcript $p = .018$) but do not survive the frozen family-wise correction. TOST on transcript-complete τ_{replay} at margin ±3pp is rejected in both models: *full transcripts realize replay materially*. The eco arm, forced to the same 300-token brutality as H-C, collapses to ≈ 0 in 7B ($-0.003\ [-0.083, +0.083]$), mirroring H-C's earlier -0.014.

6.3 Resolution

Our earlier H-C anomaly is thus identified as *sensitive to coverage/truncation*, not to representation. We deliberately state its strength precisely: the result shows transcripts *can* realize replay when their decision content is preserved; it does not isolate a clean mechanism status for coverage (multiplicity-corrected coverage contrasts are not significant). See Section 8 for the mapping to the overall strength of evidence.

7 Baselines cannot recover the decomposition

We ask whether existing evaluation signals substitute for URN oracle-randomized decomposition, by embedding three strongest neighbors *inside our own harness* with identical models and budgets.

7.1 The transplant 2×2 (Feng et al., 2026b) cannot tell replay from structure

We reproduce their architecture×content factorial structure: two write-path architectures (procedural card; raw transcript) × three content conditions (matched correct (A11 ∪ A10); near-miss; unrelated) on identical families (N=3840+3840 rollouts, model = Qwen2.5-7B). It reports matched-content effects ($+0.164$, $+0.035$ n.s.) but *by construction* cannot decompose the A11-vs-A10 legs of what counts as "matched". Our P×S split is Holm-significant in both procedural models ($+0.144$, $+0.102$), is zero on raw (-0.014), and the arch×content interaction $I_{\text{matched}} = -0.129$ is also Holm-significant—which shows: (i) representation changes the replay share, not only the total; (ii) nothing in the arch×content frame tells one that in this experiment 72% of the pooled matched gain is replay. Pair-level $P \times S$ is not a higher-resolution version of transplant factorial; it asks a question the 2 by 2 cannot define.

7.2 Embedding-based structural signal (Anonymous, 2025c): only partially predictive

Memory–task embedding similarity (frozen bge-small, our calibration view) reaches AUC 0.60 at predicting P within the trap plane and $0.35 - -0.38$ in the structural plane (chance), plus a positive sim→uplift slope significant only for 7B after Holm correction ($p_{\text{Holm}} = 0.032$). Critically it does not predict harmful flips ($p > 0.1$, odds ratios). Distilling F-MED to an embedding index thus erases both the trap direction and the clean structural estimate itself.

7.3 LLM intent-mismatch judges (Anonymous, 2026n): miscalibrated at scale

The 512-judgment blind alarm on (task, memory) pairs fires on A00/A01/A10/A11 at $100\%/93.8\%/100\%/90.6\%$ (AUC=0.508, i.e., no usable discrimination), consistent with the program-equivalence failure in Section 5.4. Family-level alarm does not predict per-family harmful-flip either (n.s.).

8 What we deliberately did not claim

1. **H-C was a formal NO-GO.** The three-form write-path minimal gate failed its frozen criterion. We did not reframe it as a positive script-form effect; we diagnosed its truncation artefact with H3 and report both.
2. **H3 was a formal NO-GO on its primary estimand.** The frozen form criterion did not reach significance; coverage is supported (directionally consistent, two models, robust simple effects) but *not confirmed* under the frozen multiplicity family. We write it as "supported-but-unresolved".
3. **TRU-Mem removed.** A transportable-utility admission learned against this oracle at minimal risk is the obvious applied sequel, but the method-side delta over Zhang et al. (2026); Anonymous (2026m;d); Cao et al. (2026) did not meet our own pre-registered differentiation standards. It is not submitted for review.

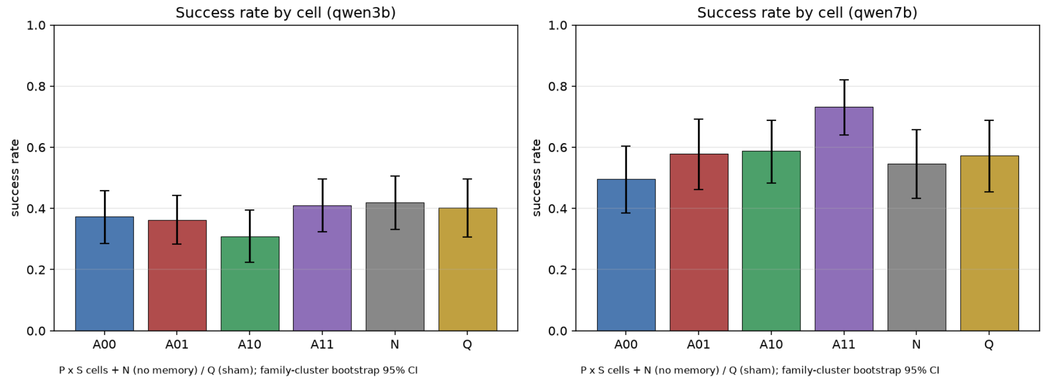

Figure 1: Stage-A cell success rates with 95% family-cluster CIs. A11 (replay-like) dominates the 7B pooled gain; the 3B pattern differs—only A11 gets near-parity with N while A10 (clean structural) falls below the unrelated control, foreshadowing the scale reversal of Section 5.2.

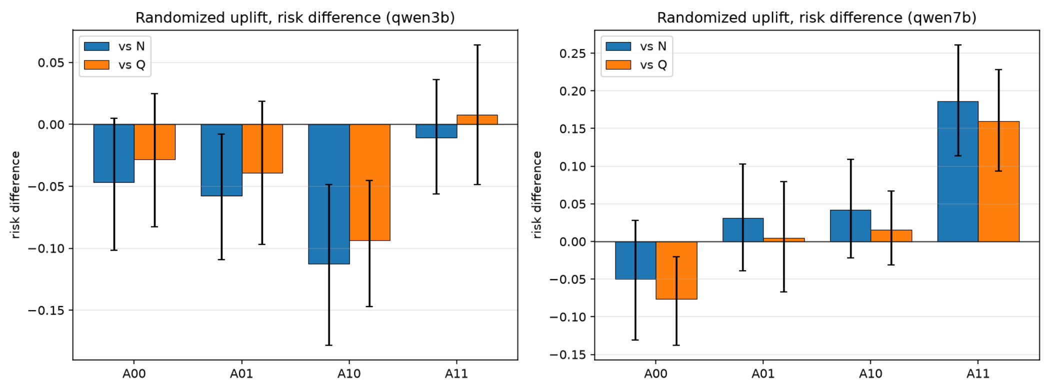

Figure 2: Risk differences relative to N and Q. The format/context effect ($Q - N$) hugs zero; the decomposition separates a small structural term from a large replay term; 3B only benefits through A11.

4. **No universal invisible-leakage assertion.** We report calibrated probe AUCs and observed strength limits, not "zero leakage proofs".

5. **Published-system profile reversal not pursued.** Two pre-registered attempts (H-C, H3) did not find significant divergence between representation forms; we do not mount a post-hoc third hunt with increased power (which would be power-chasing).

9 Discussion

9.1 What changes in the aggregate narrative

Our three main numbers are uncomfortable reading for several widely-quoted results. The pooled matched-content effect that aggregate benchmarks report is replay-dominated here (72% from the A11 leg), but the same pooled statistic would have survived two very different decompositions—a trap (A01) moving 18.8% of solvable tasks to failure for 3B, or a clean 9.2pp structural transfer for 7B. "Memory helps" is an aggregate claim that the community has been shipping where it wasn't measurable. System-level transplant experiments could not have caught this difference (§7): their factors change systems, not the semantic relation between a candidate memory and a target task.

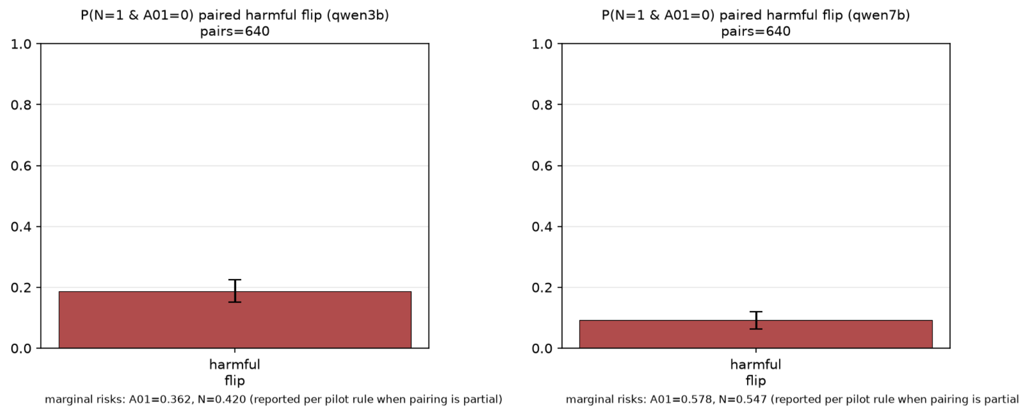

Figure 3: Paired harmful flips, A01 cell. Concentration on branches where the near-miss program semantically differs from the correct program, not a uniform effect; magnitude falls with scale.

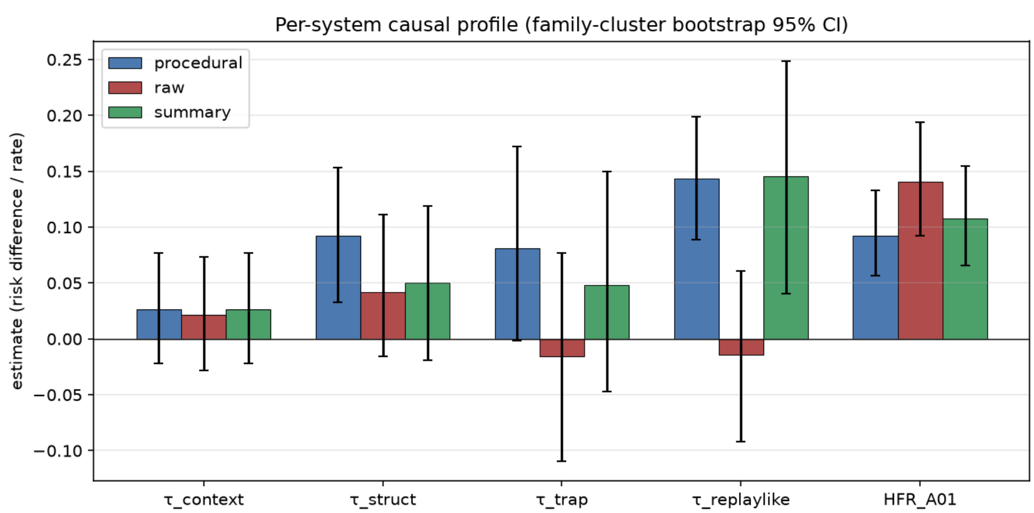

Figure 4: Form/coverage arms on fresh families (H3). Top pad: A10/A11 rates per arm; bottom pad: replay term per arm with 95% CIs. Complete transcripts realize replay materially (TOST rejects ±3pp equivalence), hard-truncated (eco) does not; form contrast is Holm-null in 8 tests.

9.2 Deployment reading

Concrete, directly-usable implications:

1. **Stop consulting aggregate accuracy** when comparing memory systems. Report the four numbers: context, clean structural, replay, and flip rate.
2. **Preserve the decision+finish of an episode** in any stored form. Our truncation results show that a 300-token cap that silently eats the decision step turns useful experience into effectively zero-signal text ($\tau_{\text{replay}} \approx 0$), even with perfect surface match.
3. **Do not trust an LLM's own equivalence check** for memory admission. We tested retrieval-grade judgment plus intent-mismatch alarms; neither recovers the label at usable strength in our environment. Program-equivalence estimation is an inference problem, not a lookup.
4. **Treat near-miss content as a fault channel**, not as style noise: paired harmful flips are model-scale-dependent (18.8% at 3B, 9.2% at 7B) and branch-conditional on program divergence.

5. **Expect structural transfer to be capability-gated**, not universal: the structural term changed sign between 3B and 7B under identical content and budget. If your fleet includes small models, measure whether structural memory *hurts* it before shipping.

9.3 LIMITATIONS

1. **Synthetic environment.** All identification is by construction in one SQLite world with programmatic predicates; transport to rich real tasks is checked only coarsely (a small ALFWorld external check; we do not claim AppWorld-scale external validity). The Tas test is whether aggregate directions repeat, not magnitudes.
2. **Narrow model coverage.** Two frozen Qwen2.5 scales drive the main inference; a Llama-3.1-8B pilot replication (same 40 families, 3840 rollouts) confirms replay-dominance (A11 leg = 79% of matched effect), the positive structural signal (7B-like, $\tau_{\text{struct}} = +0.047$ $[+0.020, +0.073]$), and similar flip magnitude (HFR=0.100). Directionally consistent, but the family count remains small, and Llama shows a small positive $\tau_{\text{context}} = +0.034$ absent in Qwen: family idiosyncrasies exist.
3. **Equivalence-class ontologys.** Our partial-order classes are stricter than unconstrained ”task similarity”; alternative operationalizations (e.g., embedding-based clustering of hidden workflows) are not evaluated and could reshuffle A10 vs A01 at the boundary.
4. **Trap-plane surface readability.** Chip-level near-miss detection from text was possible at family hold-out (AUC 0.996); we register it as a semantics of benchmark traps rather than leakage, but it limits what ”unreadable label” claims can be made on that axis.
5. **Analysis breadth.** worry: power is limited by 40/32 family clusters; decompositions at archetype level, though reported, have wide intervals and are exploratory only.

9.4 WHAT SHOULD COME NEXT

The applied sequel is a transportable-memory admission rule validated the same way—we did not run it because it would be governed against standards we cannot yet meet (Section 8). The theory sequel is a distributional-robustness treatment of transformation-shifted memory utility (Angelopoulos et al., 2024; Wu et al., 2016; Jalaldoust & Bareinboim, 2024). And the measurement sequel asks how far these estimates agree with the observational estimators people can only compute without our oracle.

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A REPRODUCIBILITY

All artifacts at `github.com/anonymous/CausalMemBench` upon de-anonymization:

- generator `pilot/generate_families.py` (pilot seed 20260807, H3 seed 20260809); sealed-vs-public split; oracle walker execution logs confirming 800/800 and 640/640 legal terminals;
- environment `pilot/env_relationalops.py`, DSL `pilot/program_dsl.py`, harness `pilot/harness.py` (vLLM 0.6.6.post1 / torch 2.5.1+cu124 on RTX A5000 24GB, fp16, temperature 0.7, top-p 0.9, max 12 steps × 512 tokens);
- rollouts JSONL: 28,416 rows (7,680 pilot + 7,680 H-C + 9,216 H3 + 3,840 Llama-8B replication; pending ALFWorld check) with per-row commit, config-hash, model, versions, GPU;
- pre-registered protocols: GATE_PROTOCOL Parts I–III dated 2026-08-07/08 before any analysis run;
- audit suite `pilot/audit/{probes,difficulty_robust,continuous_s,equivalence_audit}.py`, blind annotations, judge outputs verbatim;
- analysis: family-cluster bootstrap utilities, Holm/TOST routines, and the exact `analyze.py/json` dumps reported here;
- design registers: RESEARCH_LEDGER (pre-registration chain + external-review outcomes), §13 artifacts, construction rulings for H3 cards.

B FAMILIES AND ARCHETYPES

40 pilot families = 4 archetypes × 2 domains (crm/inv/ticket/cal) × 5 instances; 32 H3 families = 4/archetype × 8 schemas with disjoint parameters. For each family: 4 program-preserving siblings + 1 near-miss + a10 cross-domain partner + a00 unrelated partner, style counterbalanced by a 6-template Latin square, memories 200–300 Qwen-1.5B tokens measured by tokenizer, similarity buckets frozen against pilot distribution anchors.

C SECOND-FAMILY REPLICATION (COMPLETED)

Llama-3.1-8B (`NousResearch/Meta-Llama-3.1-8B-Instruct`) replicates the identical grid (3840 rollouts, same config/harness). Success rates N=.373, Q=.408, A00=.364, A01=.452, A10=.411, A11=.544. Estimates: $\tau_{\text{struct}} = +0.047\ [+0.020, +0.073]$ (7B-like, opposite 3B); $\tau_{\text{trap}} = +0.087\ [+0.043, +0.134]$; $\tau_{\text{replay}} = +0.133\ [+0.069, +0.193]$; HFR $= 0.100\ [0.075, 0.125]$; $\tau_{\text{context}} = +0.034\ [+0.010, +0.058]$ (family-specific: Qwen has ≈ 0). Replay-heavy decomposition: the A11 leg carries 79% of the matched effect, mirroring the 7B 72% share. See `pilot/llama8b/LLAMA8B_RESULTS.json`.

D EXTERNAL VALIDITY CHECK (ALFWORLD)

A coarse effect-ordering check in ALFWorld (Shridhar et al., 2021) (Qwen2.5-7B, 3 task families × 4 siblings × 4 cells × 3 seeds; memory cards frozen from expert trajectories; §A for scripts) is queued under `pilot/external/`. Reported only if it meets the same audit conventions; worse-than-N near-miss harm and replay-over-structure ordering would support external robustness of §5.